Findex Evidence on Interoperability and Inclusion

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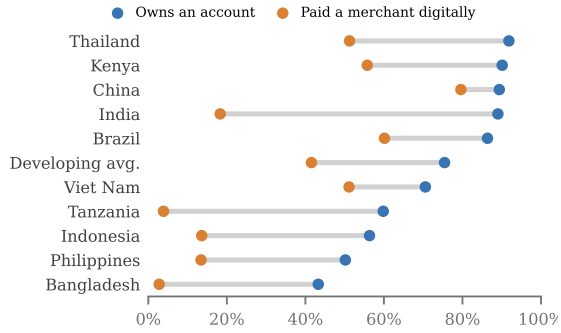
Summary: Frost, Rochet, Shin and Verdier (2025) predict that a public payment rail removes financial exclusion. In their model, inclusion means joining a payment network. The World Bank's Global Findex lets us look further. It records who owns an account, who uses it to pay merchants, and how both differ by income. Two patterns stand out. Ownership runs well ahead of merchant use, most sharply in Tanzania and India. The income gap in ownership has narrowed in Kenya, but it reopened in Brazil after 2021. Both patterns fit the paper only in part. They also show where its assumptions are weakest and where further work could go.

Findex measures more than joining

The Global Findex is the closest data we have to the paper's inclusion margin. It surveys about 145,000 adults in 141 economies every three years. In developing economies, account ownership rose from 42% in 2011 to 75% in 2024. In the model, that would be the whole story. A consumer either joins a network or stays out. Findex goes further. It records whether adults paid a merchant digitally and whether their account sat unused. It also shows how each measure varies by income, gender and phone ownership. This makes it possible to test the paper on use as well as access. It also shows how much ownership owes to policies other than payment infrastructure. Across developing economies, 43% of account owners opened their account to receive a government payment or a wage.

Chart 1. Owning an account is not paying with it

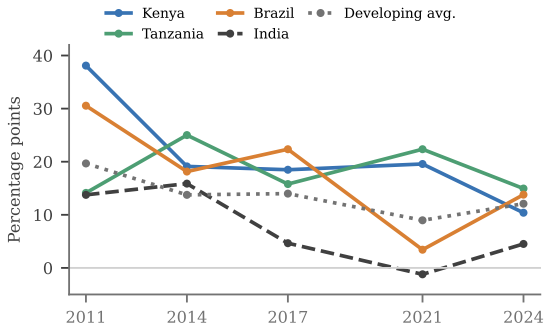
Share of adults, 2024



Source. World Bank, Global Findex Database 2025.

Chart 2. The income gap closes, then reopens

Account ownership, richest 60% minus poorest 40%



Source. World Bank, Global Findex Database 2025.

Ownership runs ahead of use

The clearest pattern in the data is a gap between owning an account and paying with it. Chart 1 shows both measures for 2024. In China, 89% of adults own an account and 80% paid a merchant digitally. In India, ownership is also 89%, but merchant use is only 18%. In Tanzania, 60% own an account and only 4% paid a merchant digitally. This is inclusion on paper, and the model cannot see it. The Tanzanian case is telling. Its mobile money providers have been fully linked for transfers since 2016. Linking consumers to each other has not brought them to merchants. That points to a weak spot in the model. The paper assumes that every merchant already accepts digital payment. Where merchants do not, consumer-side interoperability adds little to use. Part of the gap may be measurement, as Findex may miss traders paid by person-to-person transfer. Kenya offers a contrast (Chart 3). Its merchant use rose from 37% in 2021 to 56% in 2024, a period in which merchant tills were linked across networks. Tanzania stayed at 4%. The comparison is suggestive, not causal. India adds a second measure (Chart 4). In 2017, 31% of Indian adults held an inactive account. By 2024 the share had fallen to 14%. Use caught up with ownership slowly, even with UPI in place from 2016.

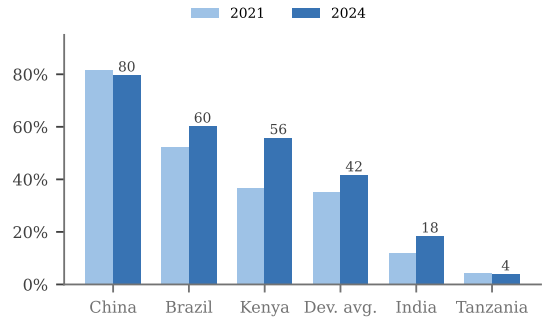
The marginal consumer is poor, and the gap moves

In the model, the consumers left out are those furthest from an intermediary. Income is the nearest observable proxy. Chart 2 shows the gap in ownership between the richest 60% and the poorest 40%. Kenya's gap fell from 38 points in 2011 to 10 points in 2024. India's has stayed below 5 points since 2017, after a government drive to open accounts. Brazil is the interesting case. Its gap fell from 22 points in 2017 to 3 points in 2021. It then reopened to 14 points in 2024. Pix launched in November 2020. Brazil also paid pandemic emergency aid into newly opened digital accounts that year. The dip and partial reversal may mean that some accounts opened to receive transfers were not kept. In the model, a rail includes the marginal consumer for good. A reversal of this

size sits awkwardly with that prediction. Across developing economies, the gap also widened, from 9 to 12 points between 2021 and 2024. Country samples are small, so movements of a few points may be noise.

Chart 3. Kenya moved, Tanzania did not

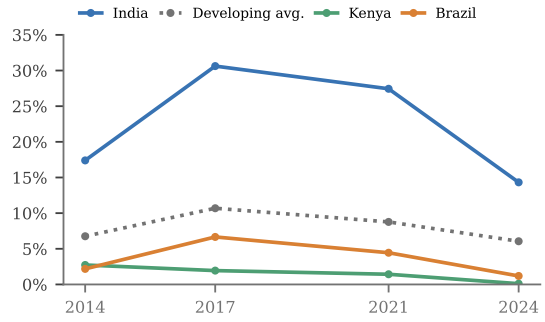
Adults who paid a merchant digitally



Source. World Bank, Global Findex Database 2025.

Chart 4. India's dormant accounts are shrinking

Adults with an inactive account



Source. World Bank, Global Findex Database 2025.

What Findex can and cannot test

Findex is strong on the consumer side and weak on everything else. It cannot see merchant acceptance, merchant fees or incumbent profits, which carry several of the paper's predictions. It cannot measure the quality of cross-network payments either. Its waves are three years apart and most reforms fall between them, so event studies need higher-frequency data. Samples of around 1,000 adults per country also limit precision. The microdata help. They allow individual-level models of who pays merchants digitally, by income, phone ownership and account type. They also record why adults opened an account, which separates transfer-driven inclusion from payment-driven inclusion. A natural design pairs the five waves with the dates and types of reform across countries. Central bank payment statistics and merchant surveys would then fill in the merchant side.

Points to explore further

Several questions follow from the data. Does merchant-side interoperability raise use more than consumer-side linking? Kenya and Tanzania offer a starting comparison. Why did Brazil's income gap reopen after 2021, and how much of the 2021 dip reflects pandemic transfers rather than Pix? Would Tanzania's merchant figure change if person-to-person payments to traders were counted? Does the fall in Indian inactive accounts track the spread of UPI, using state-level payment data? Finally, does the model need a merchant participation margin to explain these patterns? The evidence here suggests it might. Without one, the paper's rail includes consumers who have few places to pay.

Sources. Frost, Rochet, Shin and Verdier (2025), BIS Working Paper 1301. World Bank, Global Findex Database 2025, country file and report. CGAP, Tanzania Interoperability Post-Implementation Review (2018). Central Bank of Kenya, press release on mobile money interoperability (10 April 2018). Business Today Kenya and TechCabal on merchant till interoperability in Kenya. Sarkisyan (2023), Instant Payment Systems and Competition for Deposits.

This note was prepared as a writing sample. The views expressed are the author's own.